

ABHAY A



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PROFESSIONAL SUMMARY

Innovative AI Engineer specializing in computer vision, real-time video analytics, and advanced Deep Learning methodologies. Skilled in leveraging frameworks like DeepStream and Python to craft scalable, high-impact solutions for complex challenges. Renowned for a creative mindset and technical expertise, committed to shaping transformative AI applications.

PROFESSIONAL EXPERIENCE

Associate AI Engineer

Aug 2023 - Present

SafetyWhat (Facilig Intelligent Solutions Pvt Ltd) ,Kochi India

- Engineered keypoint annotation workflows for complex human pose detection, ensuring seamless model integration with high FPS performance.
- Built and deployed segmentation models for factory object detection, optimizing accuracy for industrial applications.
- Leveraged DeepStream for scalable real-time inference, integrating segmentation, keypoint, and object detection models.
- Automated YOLO annotations, achieving 90% accuracy and reducing data preparation time from a month to four days.

AI Intern

July 2023

Facilig Intelligent Solutions Pvt Ltd ,Kochi India

- Constructed an automated quality assurance pipeline for annotated datasets, achieving 92% accuracy and cutting review time by 70%, enabling rapid and reliable data preparation for complex projects.
- Streamlined workflows across diverse annotation types and tools, enhancing scalability and precision for large-scale AI initiatives.

Data Science Intern

Nov 2022 - July 2023

Luminar Technolab ,Kochi India

- Applied lean data science to real-time projects using classification, regression, and clustering across domains.
 - Utilized Hadoop, Spark, and Sqoop for data processing and created visualizations with PowerBI and Tableau.
 - Delivered end-to-end solutions, showcasing expertise from data collection to model deployment.
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EDUCATION

Kerala University, Thiruvananthapuram

Jun 2019 - May 2022

Bachelor of Science in Chemistry

National Council for Technology and Training

Nov 2022 - May 2023

Professional Development in Big Data Analytics and Data Science (Grade: A+)

Plus Two

Jun 2017 - April 2019

Govt H S S Bhoothakkulam (Biology Science)

SKILLS SUMMARY

- Languages: Python
 - Frameworks: TensorFlow, Keras, PyTorch, NumPy, Pandas, OpenCV, YOLO, ANN, CNN, RNN, Transformer Models, Image Processing, OCR, Data Analytics, Data Visualization (Matplotlib, Seaborn), OpenVINO
 - Tools: Git, Jupyter Notebook, Kafka, MongoDB, MySQL
 - Platforms: NVIDIA DeepStream, Docker, AWS, Google Colab, VS Code, LabelImg, Supervisely, CVAT, VGG Annotator, LabelMe
 - Professional Skills: Project Management, Strategic Problem-Solving, Sales and Marketing Strategy, Data Annotation, Machine Learning, Deep Learning, Programming, Data Visualization Tools, Statistics
 - Soft Skills: Effective communication, team collaboration, adaptability to dynamic environments, analytical thinking, leadership, and strategic planning.
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PROJECTS

Advanced Action Recognition in Crowded Areas

- Assembled a real-time action recognition system using YOLO, achieving 98% accuracy in identifying critical actions like waving, triggering emergency alerts to enhance safety in crowded environments.
- Created a custom dataset of 2,000+ action patterns from RTSP streams and implemented a scalable end-to-end pipeline, leveraging techniques like PCA, t-SNE, and LSTM for optimization and real-time deployment.

Advanced Hazard Identification and Segmentation in Complex Environments

- Customized a real-time segmentation system using YOLOv8 to identify and classify hazardous and non-hazardous objects (e.g., scissors, chairs), achieving 99% accuracy and enhancing safety through intelligent hazard alerts.
- Enhanced a custom annotated dataset with VGG Image Annotator, implemented a scalable pipeline, and integrated visual hazard indicators (red/green bounding boxes) for actionable insights in high-risk environments.

3D Ergonomic Assessment Using REBA and RULA Frameworks

- Constructed an AI-driven system integrating REBA and RULA frameworks with 3D posture analysis using SMPL, achieving 70% accuracy in identifying ergonomic risk levels to reduce health issues from prolonged static postures.
 - Developed a scalable risk classification pipeline with 3D mesh visualization in MeshLab, providing actionable alerts and setting the groundwork for AI-assisted workplace health management solutions.
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CERTIFICATIONS

Building Video AI Applications at the Edge on Jetson Nano

NVIDIA, Credential ID:
0d5e0dbcbe0f406abf1001daaab5fb2d (Issued Oct 2023)

Data Analytics Essentials – Cisco (Issued Jun 2023)

EXTRACURRICULAR ACTIVITIES

National Cadet Corps - India Jun 2019 - Apr 2022

Contributed to disaster and humanitarian relief activities.

National Service Scheme (NSS) Feb 2017 - Feb 2019

Engaged in community service and social work projects.